

SARVAR ABDULLAEV

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PROFILE

- AI engineer and researcher specialising in multi-agent systems, game theory and reinforcement learning. My PhD explored auction-based mechanisms for option pricing and resulted in several peer-reviewed publications.
- Co-founded an agritech startup (amudar.io) and serve as CTO where I design and deploy IoT networks for precision agriculture, develop RL models for pest monitoring and build multi-agent decision systems for greenhouses.
- 15+ years of software engineering experience, including remote roles, building scalable micro-services, cloud systems and AI pipelines. Open to remote positions in startups or tech companies as an AI engineer.

EDUCATION

King's College London (UK), PhD in Computer Science 2011–2016
Thesis on pricing options via double auctions covering agent-based modelling, automated trading and mechanism design.

Dongguk University (S. Korea), MSc in Computer Science (GPA 4.38/4.5) 2007–2009
Thesis on intelligent web event fetching; modules included applications of AI, distributed processing and software engineering.

University of Westminster (UK), BSc in Business Computing (2:1) 2003–2006

TECHNICAL EXPERIENCE

amudar.io – Agritech Startup 2020–Present
Co-founder & Chief Technology Officer Uzbekistan

- Developed an IoT ecosystem of weather stations and smart pheromone traps to support precision agriculture across Uzbekistan. Deployed more than 150 agro-weather stations and traps that provide real-time climate and pest data.
- Designed deep learning models for moth detection and counting from trap images, enabling accurate biofix detection for degree-day models used in predicting pest development stages.
- Built reinforcement-learning agents that integrate pheromone trap data with pest development cycle prediction models.
- Raised over \$190K in seed funding and established scalable cloud infrastructure for processing real-time IoT data.

cBizSoft, Inc. 2008–2023
Senior Software Engineer (Remote) USA

- Led the development of Exelare ATS, an applicant tracking system that processes more than 20K resumes daily using NLP for skill extraction and job matching.
- Integrated Exelare with external platforms (Monster, CareerBuilder, Indeed, Dice and others) and built micro-services using .NET and Angular to handle high-volume data.
- Designed intelligent document parsing and classification pipelines that reduced manual processing by 70%.

Key Projects 2018–2024
Principal Investigator Uzbekistan

- **HavoToza – Air Quality Interpolation (2022–2023):** Installed a network of ten air-quality stations across Tashkent and used traffic and meteorological data to interpolate pollutant concentrations (PM 1.0/2.5/10.0) at 10 m resolution.
- **GozanLink – Energy-Efficient Greenhouse Monitoring (2021):** Designed sensor networks that analyse solar radiation, temperature, humidity and ventilation to optimise heating in greenhouses.

ACADEMIC & CONSULTING EXPERIENCE

Inha University in Tashkent

Associate Professor

2016–Present

Uzbekistan

- Teach courses in game theory, algorithms, databases and AI fundamentals; supervise student projects in machine learning and multi-agent systems.
- Founded laboratory facilities (3D printing and VR/AR) and organised hackathons attracting more than 200 participants.

New Uzbekistan University

Adjunct Professor

2021–Present

Uzbekistan

- Designed curricula for AI/Robotics programmes and teach the Nand2Tetris course and human–computer interaction.

Consulting Projects

2018–2024

- Provided consultancy for UN and World Bank projects on remote sensing, sustainable rural development and COVID-19 response; delivered workshops on data analysis and network epidemiology.
- Developed early warning systems for pest and disease outbreaks in Ferghana valley using pre-cached maps, QGIS and GSM alerts.

TECHNICAL SKILLS

Programming	Python, C#, TypeScript/JavaScript, Java, C++, SQL, PHP
AI/ML	CrewAI, LangChain, DSPy, PyTorch, scikit-learn, OpenCV
DevOps	Azure, Docker, K8s, Ansible, Github Actions, Gitlab
Theory:	Multi-agent Systems, Game Theory, Optimization, Numerical Analysis
Sciency Tools	NetLogo, Prolog, AMPL, Gurobi, CPLEX

SELECTED PUBLICATIONS

- S. Abdullaev, P. McBurney, K. Musial-Gabrys, Pricing Options with Portfolio-holding Trading Agents in Direct Double Auction, *ECAI 2016*
- S. Abdullaev, P. McBurney, K. Musial-Gabrys, Direct Exchange Mechanisms for Option Pricing, *EUMAS 2014, Springer LNCS*
- S. Abdullaev, Trading Option Portfolios using Combinatorial Exchange, *BigDAS 2017 - Best Paper Award*
- S. Abdullaev, Ko I.S., Na Y.J., An Optimization of CDN using Efficient Load Distribution, *J. Universal Computer Science*, 2008

AWARDS & RECOGNITION

Faculty Enrichment Program Scholar (American Councils)	2025
“Shuhrat” Medal for contributions to IT education in Uzbekistan	2023
IsDB Postdoctoral Scholarship for Sustainable Agriculture	2022
Best Paper Award, BigDAS Conference	2017
Winner, TechCrunch Disrupt Hackathon (London)	2014

PROFESSIONAL DEVELOPMENT

Deep Reinforcement Learning, EEML Summer School, Warsaw	2020
ML Applications in Medicine, OxfordML Summer School	2020
Bayesian Machine Learning, DeepBayes Summer School, Moscow	2018
Entrepreneurship for Scientists, Cambridge	2012